MEDIUM · ADVANCED MATH · ANSWER KEY

Advanced Math

02

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

C

C. 16

Choice C is correct. Since the left-hand side of the given equation has a factor of $x + 9$ in both the numerator and the denominator, the solution to the given equation can be found by solving the equation x - 9 = 7. Adding 9 to both sides of this equation yields x = 16. Substituting 16 for x in the given equation yields $\frac{16 + 916 - 9}{16 + 9}$ = 7, or 7 = 7. Therefore, the solution to the given equation is 16.

Choice A is incorrect. Substituting 7 for x in the given equation yields $\frac{7 + 97 - 9}{7 + 9}$ = 7, or -2 = 7, which is false.

Choice B is incorrect. Substituting 9 for x in the given equation yields $\frac{9 + 99 - 9}{9 + 9}$ = 7, or 0 = 7, which is false.

Choice D is incorrect. Substituting 63 for x in the given equation yields $\frac{63 + 963 - 9}{63 + 9}$ = 7, or 54 = 7, which is false.

Q2

C

C. $(x+\sqrt{5})(x-\sqrt{5})$

Choice C is correct. The expression can be written as a difference of squares $x^2 - y^2$, which can be factored as $(x + y)(x - y)$. Here, $y^2 = 5$, so $y=\sqrt{5}$, and the expression therefore factors as $(x+\sqrt{5})(x-\sqrt{5})$.

Choices A and B are incorrect and may result from misunderstanding how to factor a difference of squares. Choice D is incorrect; $(x + 5)(x - 1)$ can be rewritten as $x^2 + 4x - 5$, which is not equivalent to the original expression.

Q3

18

The correct answer is 18. It's given that an egg is thrown from a rooftop and that the equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. If follows that the height, in meters, from which the egg was thrown is the value of h when $t = 0$. Substituting 0 for t in the equation $h = -4.9t^2 + 9t + 18$ yields $h0 = -4.90^2 + 90 + 18$, or $h = 18$. Therefore, according to the equation, the height, in meters, from which the egg was thrown is 18.

Q4

A

A. 1

Choice A is correct. Substituting x^2 for y in the second equation gives $2(x^2)+6=2(x+3)$. This equation can be solved as follows:

$2x^2+6=2x+6$	Apply the distributive property.
$2x^2+6-2x-6=0$	Subtract 2x and 6 from both sides of the equation.
$2x^2-2x=0$	Combine like terms.
$2x(x-1)=0$	Factor both terms on the left side of the equation by 2x.

Thus, $x=0$ and $x=1$ are the solutions to the system. Since $x>0$, only $x=1$ needs to be considered. The value of y when $x=1$ is $y=x^2=1^2=1$. Therefore, the value of xy is $(1)(1)=1$.

Choices B, C, and D are incorrect and likely result from a computational or conceptual error when solving this system of equations.

Q5

32

The correct answer is 32. The sum of the given expressions is $(-2x^2+x+31)+(3x^2+7x-8)$. Combining like terms yields $x^2+8x+23$. Based on the form of the given equation, $a=1$, $b=8$, and $c=23$. Therefore, $a+b+c=32$.

Alternate approach: Because $a+b+c$ is the value of ax^2+bx+c when $x=1$, it is possible to first make that substitution into each polynomial before adding them. When $x=1$, the first polynomial is equal to $-2+1+31=30$ and the second polynomial is equal to $3+7-8=2$. The sum of 30 and 2 is 32.

Q6

B

B. 5

Choice B is correct. It's given that the graph models the number of active projects a company was working on x months after the end of November 2012. Therefore, the value of x that corresponds to the end of November 2012 is 0. The point at which $x = 0$ is the y-intercept of the graph. It follows that the y-intercept of the graph shown is the point 0, 5. Therefore, according to the model, the predicted number of active projects the company was working on at the end of November 2012 is 5.

Choice A is incorrect. This is the value of x that corresponds to the end of November 2012, not the predicted number of active projects the company was working on at the end of November 2012.

Choice C is incorrect. This is the predicted number of active projects the company was working on 2 months after the end of November 2012.

Choice D is incorrect. This is the predicted number of active projects the company was working on 4 months after the end of November 2012.

Q7

A

A. $w = -150vx$

Choice A is correct. It's given that x is positive. Therefore, multiplying each side of the given equation by -150x yields $-150xv = w$, which is equivalent to $w = -150vx$. Thus, the equation $w = -150vx$ correctly expresses w in terms of v and x.

Choice B is incorrect. This equation is equivalent to $v = -\frac{wx}{150}$.

Choice C is incorrect. This equation is equivalent to $v = -\frac{x}{150w}$.

Choice D is incorrect. This equation is equivalent to $v = w - 150x$.

Q8

C

C. $-2.95x^2 - 7.2x + 12.16$

Choice C is correct. The first expression $(1.5x - 2.4)^2$ can be rewritten as $(1.5x - 2.4)(1.5x - 2.4)$. Applying the distributive property to this product yields $(2.25x^2 - 3.6x - 3.6x + 5.76) - (5.2x^2 - 6.4)$. This difference can be rewritten as $(2.25x^2 - 3.6x - 3.6x + 5.76) + (-1)(5.2x^2 - 6.4)$. Distributing the factor of -1 through the second expression yields $2.25x^2 - 3.6x - 3.6x + 5.76 - 5.2x^2 + 6.4$. Regrouping like terms, the expression becomes $(2.25x^2 - 5.2x^2) + (-3.6x - 3.6x) + (5.76 + 6.4)$. Combining like terms yields $-2.95x^2 - 7.2x + 12.16$.

Choices A, B, and D are incorrect and likely result from errors made when applying the distributive property or combining the resulting like terms.

Q9 C

C. 10

Choice C is correct. It's given that the function f is defined by $fx = x - 4x$. It's also given that $f5 - fa = - 15$. Substituting 5 for x in the function $fx = x - 4x$ yields $f5 = 5 - 45$ and substituting a for x in the function $fx = x - 4x$ yields $fa = a - 4a$. Therefore, $f5 = 15$ and $fa = -3a$. Substituting 15 for $f5$ and $-3a$ for fa in the equation $f5 - fa = - 15$ yields $15 - -3a = - 15$. Subtracting 15 from both sides of this equation yields $--3a = - 30$. Dividing both sides of this equation by -1 yields $-3a = 30$. By the definition of absolute value, if $-3a = 30$, then $-3a = 30$ or $-3a = -30$. Dividing both sides of each of these equations by -3 yields $a = -10$ or $a = 10$, respectively. Thus, of the given choices, a value of a that satisfies $f5 - fa = - 15$ is 10.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 D

D. $12t$

Choice D is correct. Subtracting b from each side of the given equation yields $12t = c - b$. Therefore, the expression $12t$ correctly expresses the value of $c - b$ in terms of t.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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